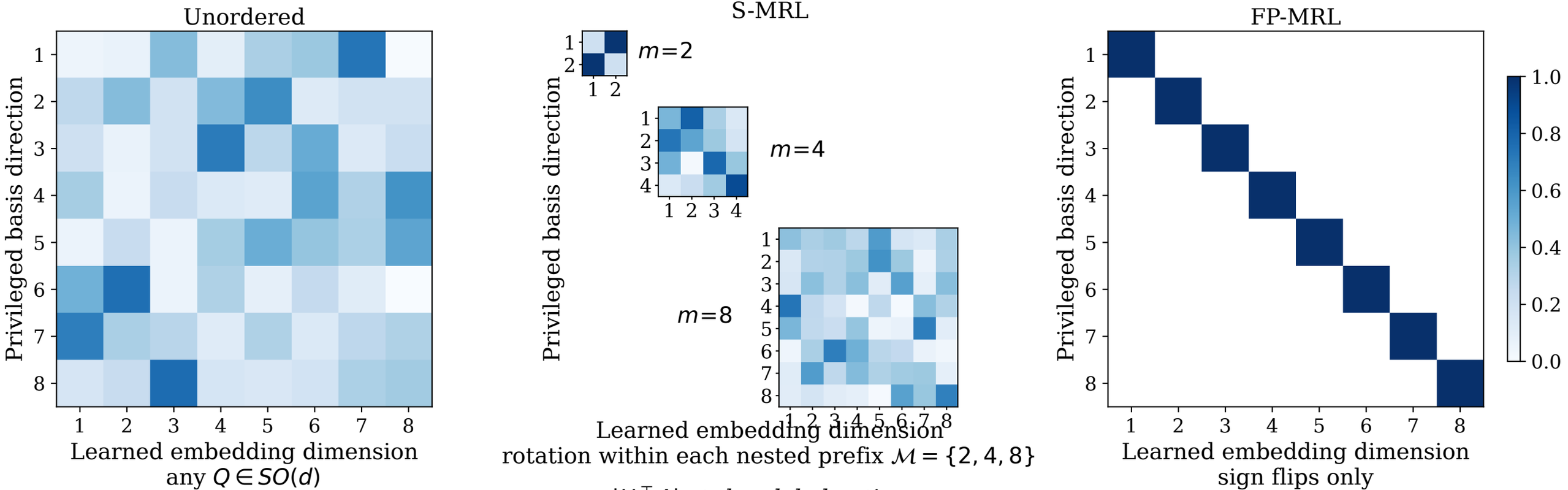
Unordered
Privileged basis direction
Learned embedding dimension
any $Q \in SO(d)$
S-MRL
$m=2$
$m=4$
$m=8$
Privileged basis direction
Learned embedding dimension
rotation within each nested prefix $\mathcal{M} = \{2, 4, 8\}$
FP-MRL
Privileged basis direction
Learned embedding dimension
sign flips only
1.0
0.8
0.6
0.4
0.2
0.0
$|U^\top A|$ *at the global optimum*